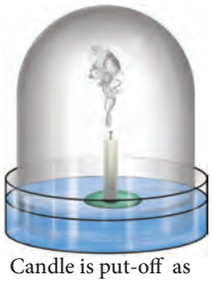
Fig-1



Candle Glows

Fig-2



Candle is put-off as there was no oxygen

Fig-3



Mouse is inserted

Fig-4



Mouse died as there was no oxygen in candle glass jar.

Fig-5



Mint plant is introduced

Fig-6



Candle continued to glow because mint provides oxygen

Fig-7



Candle glows, mouse survies with a mint plant inside.

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the candle with a glass jar. [As the bottom portion of the jar was filled with water, no air can enter or exit and hence the jar was completely sealed **(Fig-1)**]. As you would have guessed the candle flame was extinguished in a very short time. He used a magnifying glass to focus the sun rays to light the candle. Thus he tried to relight the candle many times without opening the sealed jar **(Fig-2).** The candle could not be relit. What can we make out of it?

It was clear that something in the air was being used for burning and being converted into another substance. Once the substance in the air that was aiding the burning was completely used by the burning flame and converted into another substance, the flame went out.

[Later chemist named the substance necessary for burning as oxygen and during the process of burning oxygen is converted mostly into carbon dioxide.]

Now as the jar was inside the water, Priestley could gently lift the jar and place a live mouse inside it without allowing outside air to enter the jar **(Fig-3).** Without oxygen, as you would have guessed, the mouse died **(Fig-4).** It was clear that oxygen was necessary for the survival of the mouse.

In the next step, he gently lifted the jar and placed a mint plant **(Fig-5).** (**Note**: Look at the Figure- 5; you could see that the plant is inserted into the bell jar when the jar is very much inside the water. This is done to ensure that the outside air is not entering into the bell jar.) Plant being a living thing like mouse, perhaps he thought, would die. Instead, the plant survived. After placing the mint plant, he lit up the candle and it continued to burn **(Fig-6).**

In fourth experiment, he took a jar, burned a candle and converted all oxygen into carbon dioxide. He placed a mint plant

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